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31 January 1968
Superseding
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19 October 1962

# MILITARY SPECIFICATION

# POWER SUPPLY ECU-23/A

## 1. SCOPE

1.1 This specification covers one type of unregulated, 200-amp, 28V dc, silicon diode power supply, designated ECU-23/A.

## 2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

| | |
|---|---|
| QQ-P-416 | Plating, Cadmium (Electrodeposited) |

Military

| | |
|---|---|
| MIL-M-14 | Molding Plastics and Molded Plastic Parts, Thermosetting |
| MIL-S-3151 | Sound - Level - Measuring Equipment |
| MIL-M-3171 | Magnesium Alloy, Processes for Pretreatment and Prevention of Corrosion on |
| MIL-E-5272 | Environmental Testing, Aeronautical and Associated Equipment, General Specification for |
| MIL-C-5541 | Chemical Films and Chemical Film Materials for Aluminum and Aluminum Alloys |
| MIL-G-6099 | Generators and Regulators, Air Cooled, A-C, Aircraft, General Specification for |
| MIL-I-6181 | Interference Control Requirements, Aircraft Equipment |
| MIL-S-7742 | Screw Threads, Standard, Optimum Selected Series, General Specification for |
| MIL-A-8625 | Anodic Coatings, for Aluminum and Aluminum Alloys |
| MIL-W-16878 | Wire, Electrical, Insulated, High Temperature |

STANDARDS

Military

| | |
|---|---|
| MIL-STD-100 | Engineering Drawing Practices |
| MIL-STD-130 | Identification Marking of U.S. Military Property |
| MIL-STD-143 | Specifications and Standards, Order of Precedence for the Selection of |
| MIL-STD-454 | Standard General for Electronic Equipment |
| MIL-STD-704 | Electric Power, Aircraft, Characteristics and Utilization of |
| MIL-STD-831 | Test Reports, Preparation of |
| MS3102 | Connector, Receptable, Electric, Box Mounting |

FSC 6130

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MS33586 Metals, Definition of Dissimilar

(Copies of documents require by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

## 3. REQUIREMENTS

3.1 Qualification. The power supply furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids.

3.2 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.3 Materials

3.3.1 Metals. Metals shall be of the corrosion-resistant type, unless suitably protected to resist corrosion.

3.3.1.1 Dissimilar metals. Unless suitably protected, dissimilar metals shall not be used in intimate contact. When protection is used, it shall be of such type that a low-impedance is produced for radio-frequency currents. Dissimilar metals are defined in MS33586.

3.3.2 Combustible material. Materials that will support combustion and maintain open flame after the ignition source is removed shall not be used in the power supply. High-temperature materials shall be used throughout.

3.3.3 Protective treatment. When materials are used in the construction of the power supply that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against deterioration in a manner that will in no way prevent compliance with the performance requirements specified herein. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

*3.3.4 Plastic materials. The selection and use of plastic materials shall be in accordance with MIL-STD-454 Requirement 11. The DC output terminal block shall be molded of thermosetting plastic type MAI-60 in accordance with MIL-M-14.

3.4 Design and construction

3.4.1 Design. The power supply shall be completely static with no moving parts. It shall be designed to operate from a 115/200V, 3-phase, 380 to 420 cps power source.

3.4.1.1 Current balance. The current input to each phase of the power supply shall be within 5 percent of the average value of the effective current in all phases for any condition from one-fourth to full load.

*3.4.1.2 Isolated output. The d-c output circuit shall have direct current isolation from the frame and from the a-c input circuit.

*3.4.1.3 Diodes. Paralleled diodes shall not be used.

3.4.2 Construction. The power supply shall be so constructed as to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service.

3.5 Performance. The power supply shall be capable of meeting the requirements specified herein when subjected to the following conditions:

a. Altitude - Pressure altitudes ranging from sea level to 50,000 feet



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b. High temperature - Temperatures varying uniformly from 71°C at sea level to 15°C at 30,000 feet, and 2°C at 50,000 feet

c. Low temperature - Temperature constant at -54°C from sea level to 50,000 feet

d. Humidity - Relative humidity up to 95 percent, including conditions wherein condensation takes place in the form of both water and ice

e. Vibration - Vibration incident to service use

f. Fungi - Fungus growth as encountered in tropical and semitropical climates

g. Acceleration - An applied acceleration of 10g

h. Sand dust - Sand and dust particles as encountered in desert areas

i. Salt spray - Exposure to salt sea atmosphere for a period of 50 hours

j. Shock - Shock forces of 15g as specified in 4.6.23.

*3.5.1 Output voltage. The output voltage with 200 volts, 400 cps input shall be within the limits of figure 1 for loads from 5 amperes to 200 amperes and as required to comply with 3.5.2. The output voltage shall be within the limits of 24.2 to 30.5 volts for loads from 5 to 210 amperes with input voltage variations from 190 to 210 volts, and with input frequency from 380 to 420 cps.

3.5.1.1 No load voltage. With an input voltage of 210V, a frequency of 420 cps, and a load current of 0.80 amp, the power supply d-c output voltage shall not exceed 35V.

*3.5.2 Parallel operation. Two power supplies shall be capable of sharing current to a common load with a maximum difference of 25 amperes for total load currents from 350 to 400 amperes when operating from a common a-c power source.

3.5.3 Rated load and overload

3.5.3.1 Rated load. The power supply shall be capable of delivering a continuous output of 200-amp dc.

3.5.3.2 Overload. After temperature stabilization at rated load, the power supply shall be capable of delivering 150 percent of rated load for a duration of 5 minutes under any conditions specified herein.

3.5.3.2.1 After temperature stabilization at rated load, the power supply shall be capable of delivering 250 percent of rated load for a duration of 1 minute under the conditions specified herein.

3.5.4 Short-circuit capacity. The power supply unit shall be capable of delivering a short-circuit current of 1800 amperes to 2400 amperes after temperature stabilization at rated load and under the conditions specified herein. The a-c input current for each phase required to produce the short-circuit current shall be sustained for a period of not less than that shown on figure 2.

3.5.5 Inverse voltage. The power supply shall be capable of withstanding a continuous inverse voltage of 200V and voltage transients in accordance with 4.6.10.

3.5.6 Transient input voltage. The power supply shall be capable of withstanding a transient input voltage of 230V for 10 seconds and twice rated voltage for 0.1 second.



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## VOLTAGE REGULATION
200 VOLTS, 400 CPS INPUT
25 ±10°C AMBIENT

| LOAD AMPS | VOLTAGE LIMITS | |
|---|---|---|
| | LOW | HIGH |
| 5 | 28.04 | 28.76 |
| 50 | 27.25 | 27.95 |
| 100 | 26.71 | 27.39 |
| 150 | 26.26 | 26.94 |
| 200 | 25.86 | 26.50 |
| 210 | 25.78 | 26.42 |

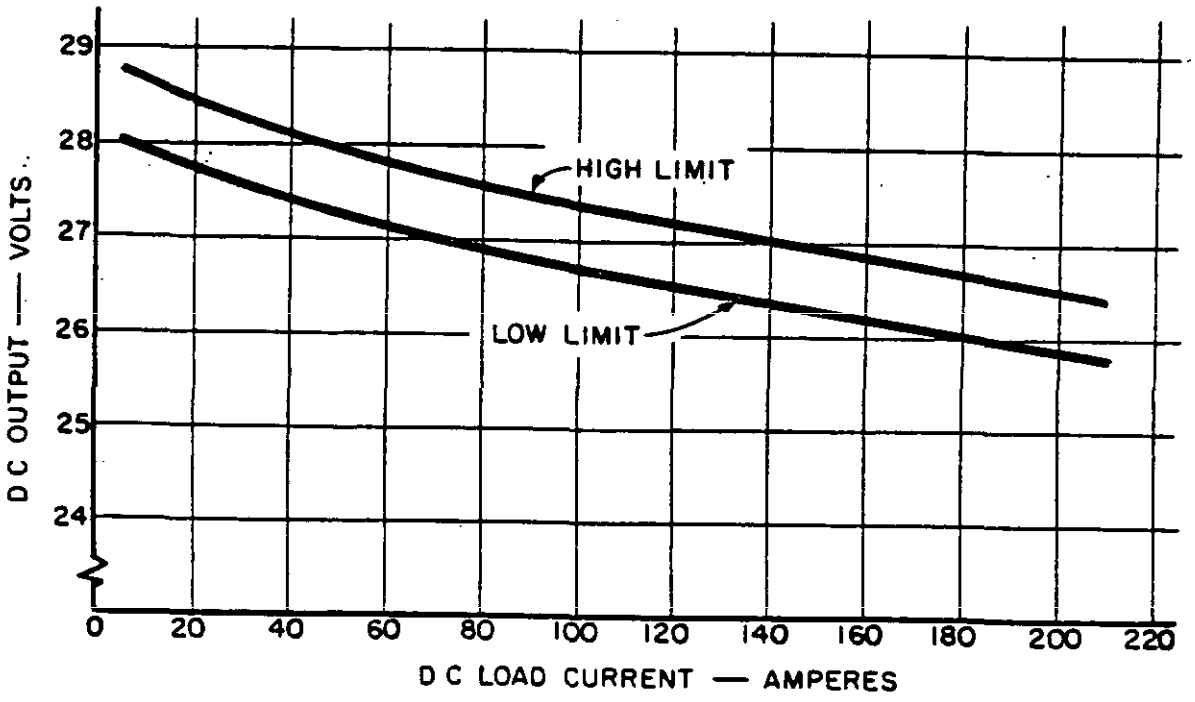



*FIGURE 1. Output Voltage



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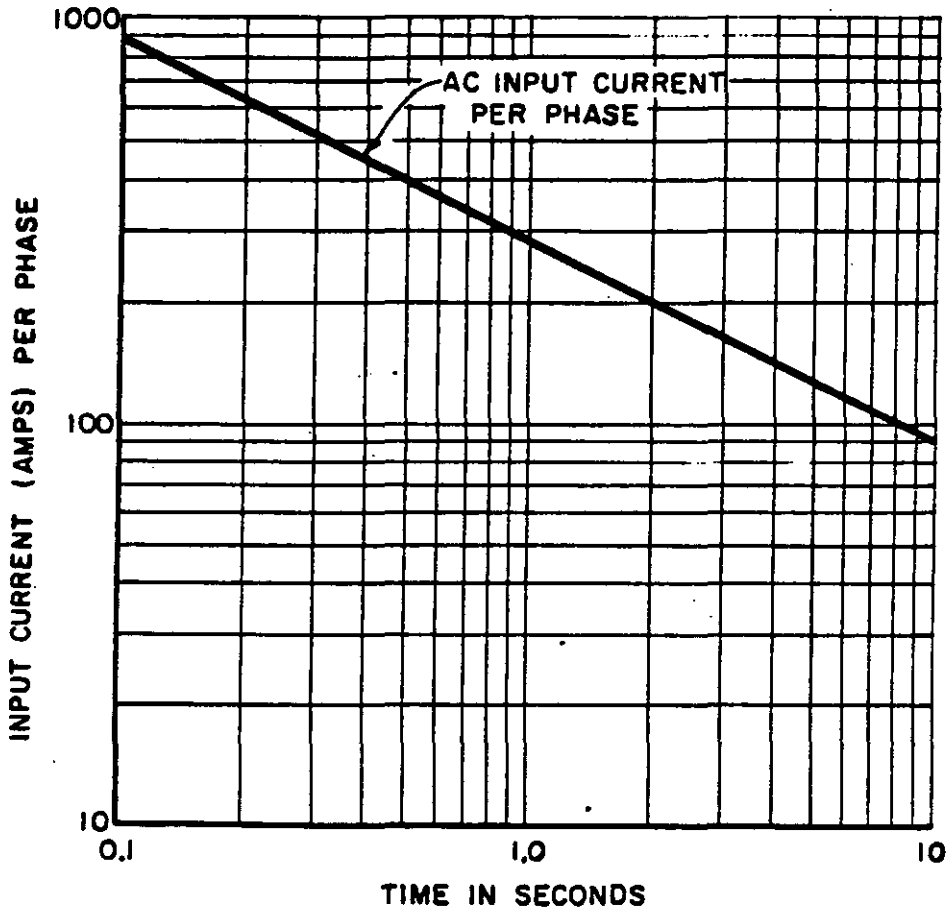



FIGURE 2. Short Circuit Rating



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3.5.7 Input factor. The input power factor shall be not less than 90 percent from one-half to full rated load.

3.5.8 Efficiency. The overall efficiency shall be not less than 85 percent from one-half to full rated load.

3.5.9 Ripple voltage. The ripple voltage shall not exceed 1.1 volt peak to peak for all loads up to full load.

3.5.10 Ripple frequency. The frequency components of the ripple shall not exceed the limits specified in MIL-STD-704.

3.5.11 Acoustical noise. The acoustical noise output of the power supply, when measured at a distance of 1 foot, shall not exceed 67 db.

3.5.12 Radio interference. The power supply shall meet the radiated interference requirements of MIL-I-6181. The maximum limit for broadband and pulsed continuous wave conducted interference shall be within the limits shown on figure entitled "Broadband and pulsed CW conducted interference limits using stabilization network" in MIL-I-6181, except that the slope of the curve shown on this figure shall be modified so that it starts at 0.1 mc, 92.5 db above 1 microvolt per mc and joins the horizontal portion of the curve at 2 mc, 81 db above 1 microvolt per mc.

3.5.13 Service life. The power supply shall have a service life of 10,000 hours without maintenance under the following load cycles:

a. 1,000 hours at 25 percent load

b. 7,000 hours at 50 percent load

c. 2,000 hours at 100 percent load.

3.5.14 Operational loads. The power supply shall supply the loads specified herein when mounted with the base down, in a horizontal plane, and with the still-air volume around the unit equal to three times the volume of the power supply. The maximum external frame temperature at rated load under any condition shall not exceed 149°C.

3.5.15 Dielectric strength. The power supply shall withstand application of 1,500V rms commercial frequency between the a-c input terminals and the frame and between the d-c output terminals and the frame for a period of 1 minute.

3.5.16 Operating positions. The power supply shall be capable of operating at any position from a normal base-down, horizontal-plane position, to a maximum tilt of 16° from this position. It shall be capable of satisfactory operation in the above tilted position for a period of 1 hour and in any position for a period of 5 minutes.

3.5.17 Threaded terminals. Terminals shall be as shown on figure 3. They shall withstand static values of 115 pounds pull and 225 pound-inches of torque in both directions for 1 minute without distortion, turning or loosening.

3.5.17.1 Terminal configuration. The output terminal studs shall be designed so that the primary current paths shall be from connecting cable terminal lugs to receiving cable terminal lugs through a brass sleeve or stud shoulder. The clamping force to hold the lugs together or against the sleeve or shoulder shall be applied through the nut and stud assembly. Plastics shall not be used in the compression stackup of lugs, washers and nuts.



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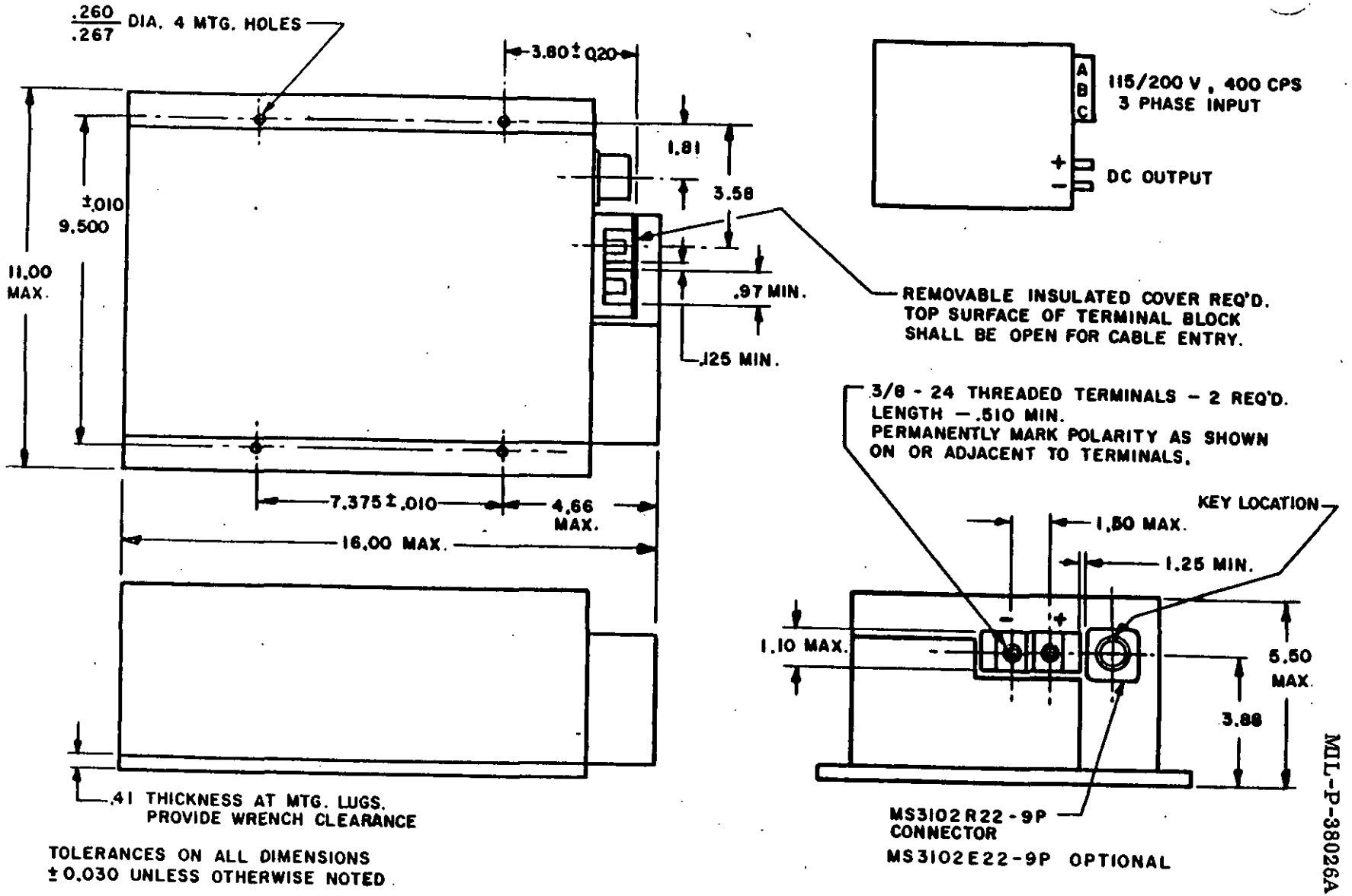



FIGURE 3. Outline Drawing



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3.6 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part numbers and changes thereto.

3.7 Wiring. The power supply shall be wired in accordance with MIL-STD-454, Requirement 9. External connections shall be as shown on figure 3. Wire shall be in accordance with MIL-W-16878, type EE.

3.7.1 Wiring diagram. A wiring diagram of the power supply shall be permanently attached to the unit.

3.8 Screw threads. All screw threads shall be in accordance with MIL-S-7742.

3.9 Outline dimensions. Outline dimensions shall be in accordance with figure 3.

3.10 Weight. The weight of the power supply shall not exceed 34.5 pounds.

3.11 Finishes and protective coatings

3.11.1 Aluminum alloy parts. Whenever practicable, aluminum alloy parts shall be treated in accordance with MIL-C-5541 or MIL-A-8625. The aluminum oxide formed by MIL-A-8625 treatment shall be removed from the actual contact area of all surfaces required to act as a path for electric current, and from local areas under screws, nuts, et cetera, used for assembly or mounting purposes, to provide adequate bonding connections.

3.11.2 Magnesium alloy parts. Whenever practicable, magnesium alloy parts shall be surface treated in accordance with MIL-M-3171 to provide protection against corrosion.

3.11.3 Cadmium plating. Cadmium plating shall be in accordance with QQ-P-416, and of a type and class adequate to achieve the degree of protection required.

3.12 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.13 Workmanship. The power supply shall be fabricated in a thoroughly workmanlike manner. Particular attention shall be given to accuracy of dimensions; thoroughness of soldering, welding, brazing, wiring, and riveting, tightness of assembly screws, bolts, and nuts; marking of parts and assemblies; and freedom of parts from defects, burrs, and sharp edges.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of power supplies shall be classified as follows:

a. Qualification tests

b. Acceptance tests.



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4.3 Test conditions. Unless otherwise specified, the following test conditions shall apply:

a. Ambient temperature: 25° ±10°C

b. Altitude: Atmospheric pressure (approximately 29.92 inches Hg)

c. Input power: 115/200V, 400 cps

d. Voltage measurement: At terminals of the power supply

e. Location of load: Such that it will not appreciably affect the ambient temperature

*f. No-load current: 0.80 amperes maximum

*g. Frame ground: The frame of the unit shall be externally connected to the neutral of the a-c supply and to the negative d-c terminal.

*4.3.1 Test enclosure. The tests specified in 4.3.1.1 shall be conducted with the power supply mounted with base down, in a horizontal plane, and in an enclosure as shown on figure 4. The enclosure shall have four times the volume of the power supply. The enclosure shall be dimensionally proportional to the unit and shall be constructed of low heat conductance material. It shall have an internal reflecting surface of aluminum foil. A 1-1/2-inch wide opening shall be made around the enclosure below the top and also around the base. The power supply shall be centrally located 1-1/2-inches above the bottom of the enclosure on insulated mounts to prevent conductive heat flow. The maximum external frame temperature at rated load during any condition shall not exceed 149°C. No cooling airflow other than that of a normally ventilated room shall be provided.

4.3.1.1 Enclosure tests. The following tests shall be conducted within the enclosure specified in 4.3.1:

a. Output voltage

b. Overload

c. Input power factor

d. Efficiency

e. Operating position

f. High temperature

g. Altitude

4.4 Qualification testing

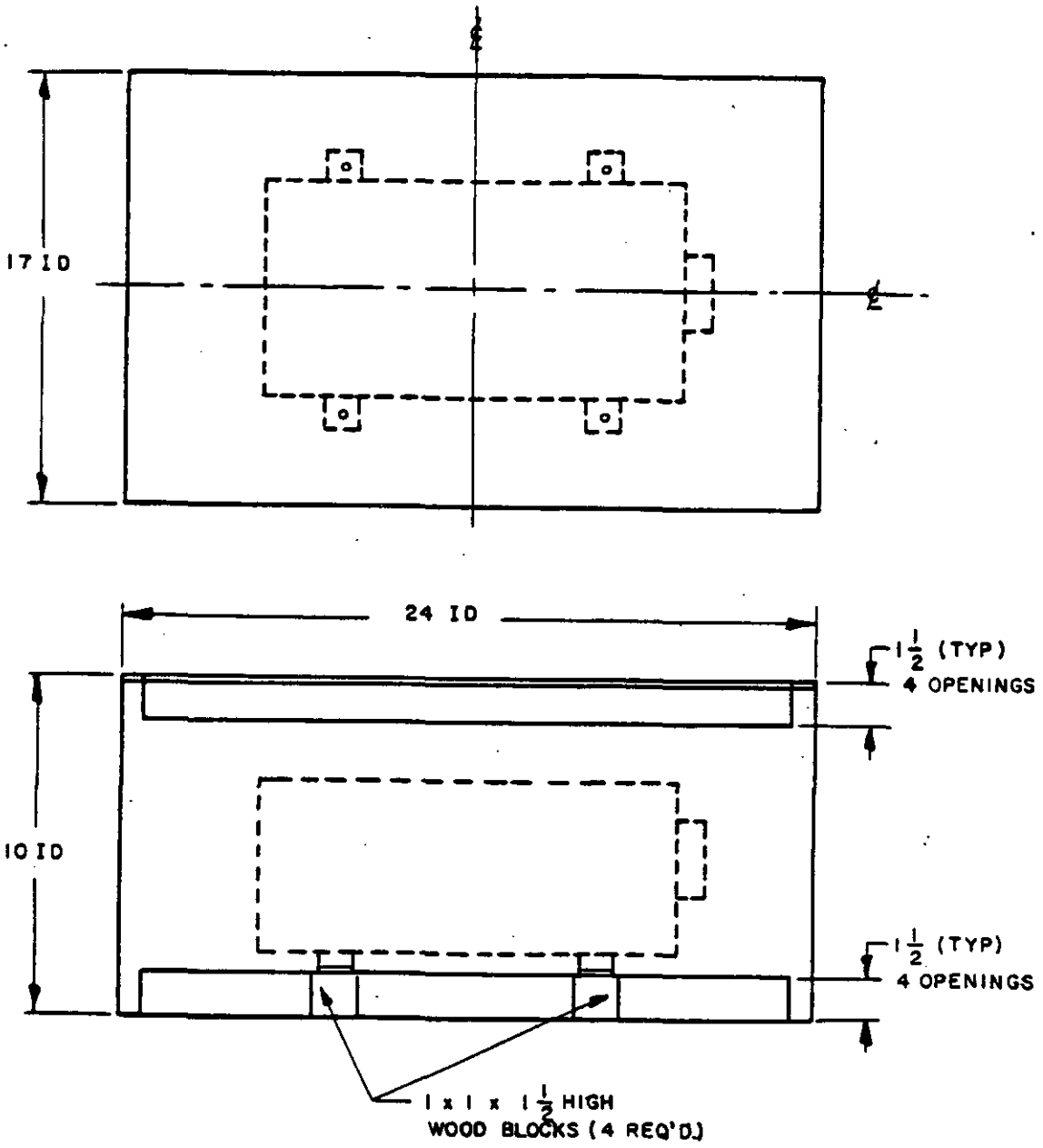
4.4.1 Test samples. The test samples shall consist of three power supplies representative of the production equipment. The samples shall be identified with the manufacturer's part number and such other information as required by the procuring activity. The power supplies shall be instrumented to permit the measurement of critical component temperatures. The instrumentation shall not appreciably change the electrical or thermal characteristics of the unit.

4.4.2 Test report. When the tests are conducted at a location other than the laboratory of the procuring activity, three copies of a test report in accordance with MIL-STD-831 shall be furnished to that activity.

4.4.3 Qualification tests. The qualification tests shall consist of all the tests specified under 4.6, conducted on the samples and in the order listed in table I.



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MATERIAL:
INSULATING BOARD,
ALUMINUM FOIL
INSIDE SURFACES

FIGURE 4. Test Enclosure



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TABLE I
Qualification Testing

| Test | Paragraph | Samples | | |
|---|---|---|---|---|
| | | 1 | 2 | 3 |
| Examination of product | 4.6.1 | x | x | x |
| Input current balance | 4.6.2 | x | x | |
| Output voltage | 4.6.3 (a thru e) | x | x | |
| | 4.6.3 (a and b only) | | | x |
| Overload | 4.6.4a | x | x | |
| | 4.6.4b | x | x | |
| | 4.6.4c | | | x |
| Dielectric strength | 4.6.5 | | | x |
| Efficiency | 4.6.6 | x | x | x |
| Ripple voltage | 4.6.7 | x | x | x |
| Short circuit capacity | 4.6.8 | x | x | |
| Parallel operation | 4.6.9 | x | x | x |
| Inverse voltage | 4.6.10 | x | x | |
| Transient input voltage | 4.6.11 | x | x | |
| Input power factor | 4.6.12 | x | x | x |
| Acoustical noise | 4.6.13 | x | x | x |
| Altitude | 4.6.14 | x | | |
| High temperature | 4.6.15 | x | | |
| Low temperature | 4.6.16 | x | | |
| Humidity | 4.6.17 | x | | |
| Vibration | 4.6.18 | | x | |
| Fungus resistance | 4.6.19 | | x | |
| Acceleration | 4.6.20 | x | x | |
| Sand and dust | 4.6.21 | | x | |
| Salt spray | 4.6.22 | | x | |
| Shock | 4.6.23 | | x | |
| Radio interference | 4.6.24 | x | x | |
| Operating position | 4.6.25 | x | x | |
| Life | 4.6.26 | | | x |
| Terminal strength | 4.6.27 | x | x | x |



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4.5 Acceptance tests. Acceptance tests shall consist of the individual tests.

4.5.1 Individual tests. Each power supply unit shall be subjected to the following tests as described under "test methods," except that temperature readings on the various component parts shall not be taken:

a. Examination of product

b. Input current

c. Output voltage (room temperature only) conducted for the full 200V, 400 cycle curve (0, 1/4, 1/2, 3/4, full load) plus no load at 210V and full load at 190V

d. Overload

e. Dielectric strength

f. Efficiency

g. Ripple voltage

h. Short circuit capacity (one test only)

i. Acoustical noise.

4.6 Test methods

4.6.1 Examination of product. The power supply shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, construction, weight, and dimensions.

4.6.2 Input current balance. For this test, the current input in each phase of the power supply shall be measured at one-fourth, one-half, and full rated load. For each load condition, the current shall be within 5 percent of the average value of the effective current in all phases.

4.6.3 Output voltage. Following a 30-minute warmup at rated load the output voltage shall remain within the limits specified in 3.5.1.

a. Output voltage at ambient temperature (25° ±10°C)

(1) At 190V, 380 cps a-c input, the load shall be varied from no load to full load. The load shall be suddenly removed and applied for each observation. The output voltage shall be recorded for no-load, 5, 50, 100, 150, 200, and 210 amperes

(2) At 200V, 400 cps a-c input the preceding test shall be repeated

(3) At 210V, 380 cps a-c input the preceding tests shall be repeated.

b. With the frequency at 420 cps, part a (1) and a (3), shall be repeated at 25° ±10°C

c. Parts a and b shall be repeated at a minimum ambient temperature of -54°C. The power supply shall be placed in the test chamber and cooled to and maintained at the minimum temperature until stabilization of the equipment is reached. While at this temperature, the unit shall be operated and the prescribed tests run

d. Parts a and b shall be repeated at a maximum ambient temperature of +71°C. The power supply shall be placed in the test chamber and the internal temperature of the chamber raised to a temperature of 71°C with an internal relative humidity of not more than 5 percent. The unit shall be maintained at this temperature until stabilization of the equipment is reached. While still at this temperature, it shall be operated and the prescribed tests conducted

e. Parts a and b shall be repeated at room ambient temperature to determine that the



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high and low temperatures did not adversely affect the power supply.

*4.6.4 Overload. Overload cycles shall be conducted as follows:

a. The power supply shall be operated for 7 hours with an input voltage and frequency of 200V 400 cps. During this period, the power supply shall be subjected to resistive load cycling. One cycle shall consist of 25 minutes at rated load followed by 5 minutes at 300 amperes load. Operation shall be satisfactory and the temperature of the components shall not exceed their maximum ratings

b. The power supply shall be operated for 7 hours with an input voltage and frequency of 200V 400 cps. During this period, the power supply shall be subjected to resistive load cycling. One cycle shall consist of 25 minutes at rated load followed by 1 minute at 500 ampere load. Operation shall be satisfactory and the temperature of the components shall not exceed their maximum ratings. Under this overload condition, the voltage shall be not less than 23V

c. For acceptance tests only, the power supply shall be operated for 1 cycle at the 300- and 500-amp overload conditions.

4.6.5 Dielectric strength. A potential of 1,500V rms commercial frequency shall be applied for 1 minute between the a-c input terminals, short-circuited together, and the frame with all grounded connections removed. The test shall be repeated with a potential of 1,500V rms commercial frequency applied between the d-c output terminals, short-circuited together, and the frame with all grounded connections removed. No breakdown of the insulation shall occur as a result of this test.

4.6.6 Efficiency. With the input adjusted at 200V 400 cps, the full-load efficiency shall be measured after an initial 30-minute warmup period. The load shall then be reduced to 50 percent and the efficiency measured again. The efficiency shall be not less than 85 percent from one-half to full rated load.

*4.6.7 Ripple voltage. The ripple voltage and frequency in the d-c output of the power supply shall be measured at no-load, one-fourth, one-half, three-fourths, and full rated resistive loads. The maximum value of the peak-to-peak ripple voltage measured for these loads shall not exceed 1.1 volt. Qualification test measurements shall be made with a cathode ray oscilloscope having an accuracy of 2 percent or better. Acceptance test measurements shall be made with a cathode ray oscilloscope having an accuracy of 2 percent or better, or with a 4 uf capacitor in series with a peak-to-peak reading vacuum tube voltmeter (Measurements Corporation Model 67, or equal) having an accuracy of 2 percent or better. If the vacuum tube voltmeter is utilized, it shall be so marked that the maximum allowable ripple voltage value does not exceed that which would be read on the oscilloscope. The frequency components of the ripple shall be within the limits specified in MIL-STD-704 when measured as conducted interference using procedures outlined in MIL-I-6181. The power supply unit shall receive power during this test from an unloaded 3-phase 208/120V 400-cps a-c generator of 40 kva capacity or larger which conforms electrically to MIL-G-6099. An alternate test for ripple voltage will be to use a 3-phase power source of 4.6V (rms) line to neutral for each harmonic frequency up to the 20th harmonic of the fundamental frequency. The peak-to-peak ripple voltage at the output terminals shall not exceed 1V for any of the harmonic frequencies tested.



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*4.6.8 Short circuit capacity. The power supply unit shall be connected to a 200V 400 cps power source and operated at rated load until the temperature stabilizes. The output terminals shall then be shorted together through a resistance of 0.0035 to 0.0040 ohms. Within 0.12 second after application of the short circuit the a-c input voltage at the terminals of the unit shall be a minimum of 180 volts. The output current shall be 1800 amperes minimum to 2400 amperes maximum. The input current shall be sustained for a period of time no less than that shown in figure 2. (Example: If the input current during the short circuit is 200 amperes, it shall be sustained for no less than 2.0 seconds). The input current value shall be that of the phase carrying the least current. The short circuit test shall be performed four times and no damage shall occur. There shall be no diode failures.

*4.6.9 Parallel operation. The parallel operation test shall be performed as follows:

a. Two power supplies shall be connected to a common load. Load division shall be verified at 350 and 400 amps. The input voltages shall be adjusted to 190, 200, and 210V at each load point. The input frequencies shall be adjusted to 380, 400, and 420 cps at each input voltage value at each load point. Load division shall be no greater than that specified in 3.5.2

b. Two power supplies shall be connected to a common load and the load adjusted to one-half the combined rating of the two units. One unit shall then be de-energized and the output voltage of the remaining unit shall remain within the limits specified in 3.5.1

4.6.10 Inverse voltage

a. A 200V inverse voltage shall be applied to the output terminals of the power supply for a period of 7 hours. No part of the power supply shall be damaged or fail to function as a result of this test. Leakage current at the end of the test shall not exceed the initial leakage by more than 5 percent. The temperature of the power supply may be stabilized at the initial point of the test and shall be approximately the same as the stabilized temperature of the unit at the end of the test. Capacitors may be disconnected from the circuit for this test.

b. The power supply shall be subjected to the transient voltages shown in table II by the discharge of a capacitor into the output terminals. The unit shall not be connected to any load or power source during this test and all capacitors and components shall be connected.

TABLE II

Transient Voltages and Capacitors

| Voltage | Capacitor |
|---|---|
| -600V (opposite polarity to the bus) | Capacitor size to release energy equivalent to 600V 350 usec transient. |
| +400V (normal bus polarity) | Capacitor size to release energy equivalent to 400V 180 usec transient. |

Each voltage shall be applied 20 times within a period of 5 minutes. A transient voltage source resistance of 50 ±5 ohms shall be included in the test circuit. The power supply shall show no signs of damage; it shall be tested in accordance with and shall meet the tests specified in 4.6.3a, 4.6.3b, and 4.6.7 after completion of the transient voltage tests.



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4.6.11 Transient input voltage. With no d-c load applied, the power supply shall be subjected to one input voltage transient of 230 volts for a minimum of 10 seconds and one input voltage transient of 400 volts for a minimum of 0.10 second. No damage shall result. Full load output voltage and ripple voltage shall be within applicable limits following the test.

4.6.12 Input power factor. With the input adjusted to 200V 400 cps, the full-load power factor shall be measured after an initial 30-minute warmup period. The load shall then be reduced to 50 percent of the rated load and the power factor measured again. The power factor shall not be less than 90 percent from one-half to full rated load.

4.6.13 Acoustical noise. The acoustical noise of each unit shall be measured 1 foot from each of six sides (including top, bottom, and sides) where the noise level is highest. The noise limit is 67 db measured by an integrating type noise meter using the "B" scale or the 70 db design objective scale defined by figure entitled "Over-all free-field frequency responses and tolerances" of MIL-S-3151. The noise level instrumentation (micro-phone-filter-amplifier system) tolerance characteristics shall conform to figure 5. The instrumentation shall be calibrated throughout the frequency range daily, prior to usage. The calibrating devices or standards used shall be directly traceable to the National Bureau of Standards. The unit manufacturer may establish, subject to procuring activity approval, a simplified test procedure for production unit acceptance tests. Such a procedure must be substantiated by data from at least 20 production units and include a random sampling test requirement (measure all sides) for two or more units. Continuous recalibration of the noise measuring equipment will be required for such a procedure.

*4.6.14 Altitude. The power supply shall be subjected to an altitude test for 6 hours at rated load in accordance with procedure VI, condition C, of MIL-E-5272. During the final hour, the power supply output voltage shall be recorded and shall be within the tolerance specified in 3.5.1. The maximum external frame temperature shall be recorded.

*4.6.15 High temperature. The power supply shall be subjected to a high temperature test at rated load in accordance with procedure II of MIL-E-5272. During the final hour, the power supply output voltage shall be recorded and shall be within the tolerance specified in 3.5.1. The maximum external frame temperature shall be recorded.

4.6.16 Low temperature. The power supply shall be subjected to a low temperature test in accordance with procedure II of MIL-E-5272. During the final hour, the power supply output voltage shall be recorded and should be within the tolerance specified in 3.5.1.

4.6.17 Humidity. The power supply shall be subjected to a humidity test in accordance with procedure I of MIL-E-5272. At the end of this test, the power supply shall show no signs of excessive corrosion of metal parts. The output voltage shall be recorded and shall be within the tolerance specified in 3.5.1.

4.6.18 Vibration. The power supply shall be subjected to a vibration test in accordance with procedure XII of MIL-E-5272 at room temperature only. After completion of this test, the power supply shall show no signs of damage or loosening of



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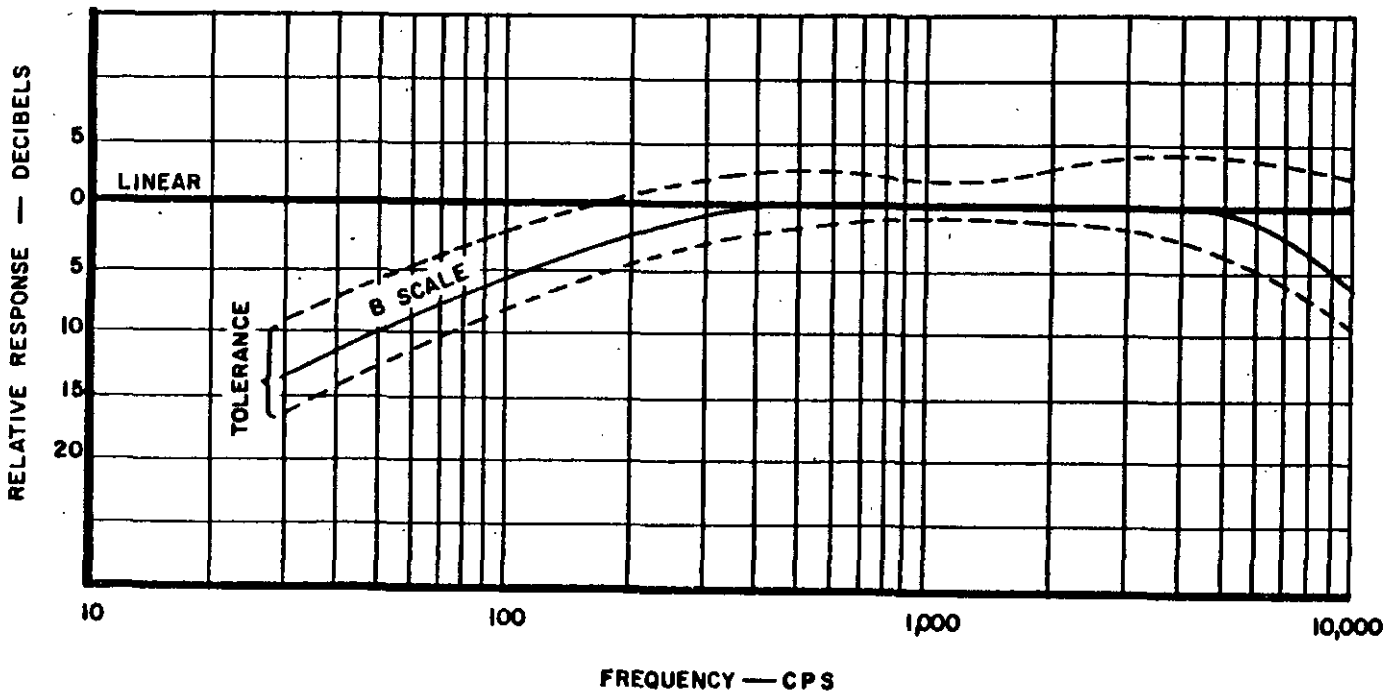



FIGURE 5. Noise Measuring System Relative Response Curve

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parts. The output voltage shall be recorded and shall be within the tolerance specified in 3.5.1.

4.6.19 Fungus resistance. The power supply shall be subjected to a fungus resistance test in accordance with procedure I of MIL-E-5272. After completion of this test, the power supply shall operate satisfactorily. The output voltage shall be recorded and shall be within the tolerance specified in 3.5.1.

4.6.20 Acceleration. The power supply shall be subjected to an acceleration test in accordance with procedure III of MIL-E-5272. The applied acceleration shall be 10g. No damage shall occur as a result of this test.

4.6.21 Sand and dust. The power supply shall be subjected to a sand and dust test in accordance with procedure I, part I of MIL-E-5272. After completion of this test, the power supply shall operate satisfactorily. The output voltage shall be recorded and shall be within the tolerance specified in 3.5.1.

4.6.22 Salt spray. The power supply shall be subjected to a salt spray test in accordance with procedure I of MIL-E-5272. At the end of this test, the power supply shall show no signs of excessive corrosion of metal parts. The output voltage shall be recorded and shall be within the tolerance specified in 3.5.1.

4.6.23 Shock. The power supply shall be subjected to a shock test in accordance with procedure V of MIL-E-5272. After completion of this test, the power supply shall show no signs of damage or loosening of parts. The output voltage shall be recorded and shall be within the tolerance specified in 3.5.1.

4.6.24 Radio interference. The power supply shall be subjected to a radio interference test in accordance with the radiated portion, and the broadband and pulsed continuous-wave conducted interference portion of MIL-I-6181 except as modified by 3.5.12. Both conductive and radiated radio interference levels shall be measured at 10-, 100-, and 200-amp resistive loads. The band width for the 100 to 150 kc frequency range shall be measured with a Stoddard interference instrument, part number NM10A, or equal.

*4.6.25 Operating positions. Following a 30-minute warmup period at rated load the power supply shall be operated at rated load in the following positions:

a. With the base tilted 16 degrees from the horizontal in each of 4 directions for a period of 1 hour in each position.

b. With the base tilted 90 degrees from the horizontal in each of 4 directions for a period of 5 minutes in each position.

c. For each position the temperatures of all components shall be within their maximum ratings and output voltage shall comply with 3.5.1.

4.6.26 Life. The life test shall consist of 1,000 hours of operation at rated load. The ambient temperature for this test shall be such that the maximum external frame temperature is equal to the maximum external frame temperature recorded during the high temperature test specified in 4.6.15. At the end of each 250 hours of operation, the power supply shall satisfactorily meet the short circuit requirements of 4.6.8 (once only). At the completion of 1,000 hours, the power supply shall be subjected to and shall withstand the tests specified in 4.6.3, 4.6.7, 4.6.10 (15 minutes only), and 4.6.13.



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4.6.27 Terminal strength. The terminals of the power supply shall be subjected to and shall withstand, without damage, the specific values of static pull and torque specified in 3.5.17. Distortion or loosening of terminals shall constitute failure.

## 5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with instructions of the procuring activity.

## 6. NOTES

6.1 Intended use. The type ECU-23/A power supply covered by this specification is intended for use in 200V 3-phase 400-cycle a-c systems to provide a 28V d-c power source.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Selection of applicable levels of packaging and packing.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Aeronautical Systems Division, Attn: ASNJP-10, Wright-Patterson Air Force Base, Ohio 45433 and information pertaining to qualification of products may be obtained from that activity.

6.4 The margins of this specification are marked with an asterisk to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

| | |
|---|---|
| Custodian: | Preparing activity: |
| Air Force - 11 | Air Force - 11 |
| Review activities: | Project No. 6130- |
| Air Force - 11, 85 | F059 |



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# SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4063, WHICH IS OBSOLETE C-0279

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